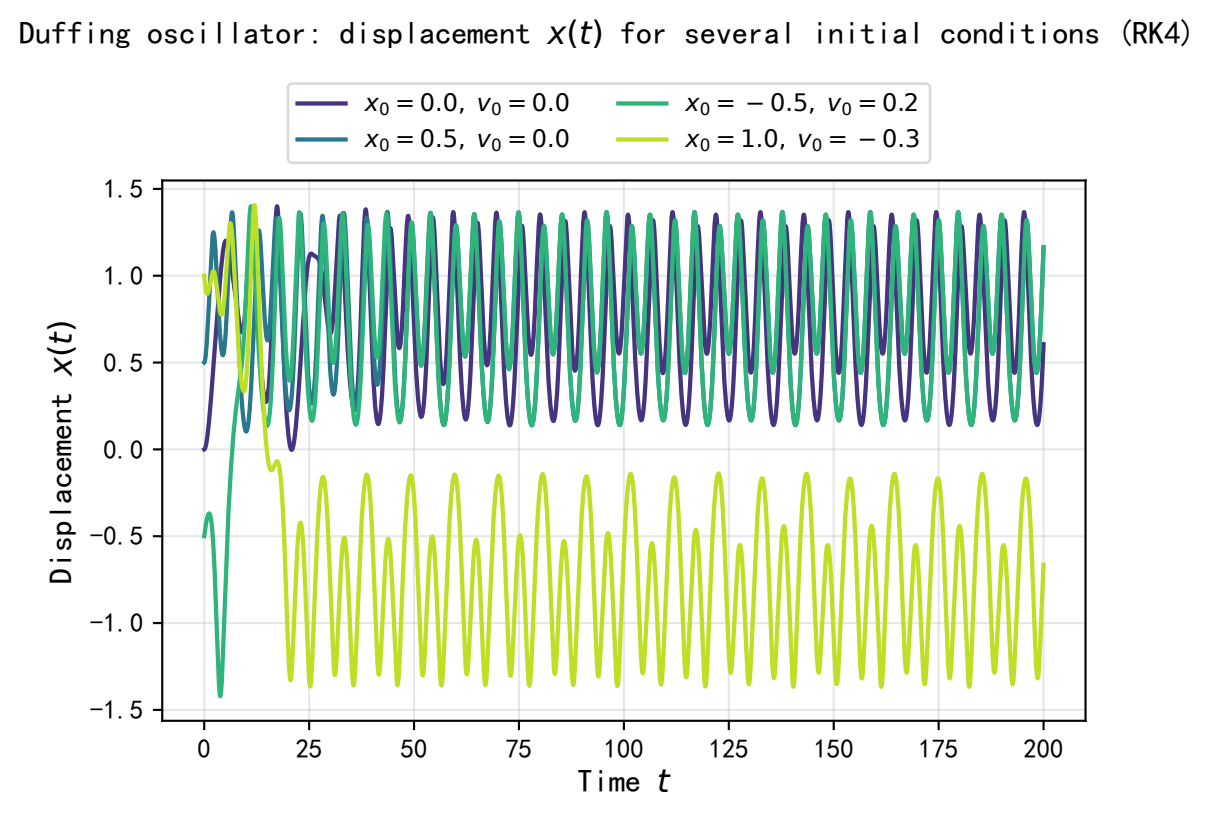
Duffing oscillator: displacement $x(t)$ for several initial conditions (RK4)
$x_0 = 0.0,\ v_0 = 0.0$
$x_0 = 0.5,\ v_0 = 0.0$
$x_0 = -0.5,\ v_0 = 0.2$
$x_0 = 1.0,\ v_0 = -0.3$
Displacement $x(t)$
Time t
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
25
50
75
100
125
150
175
200